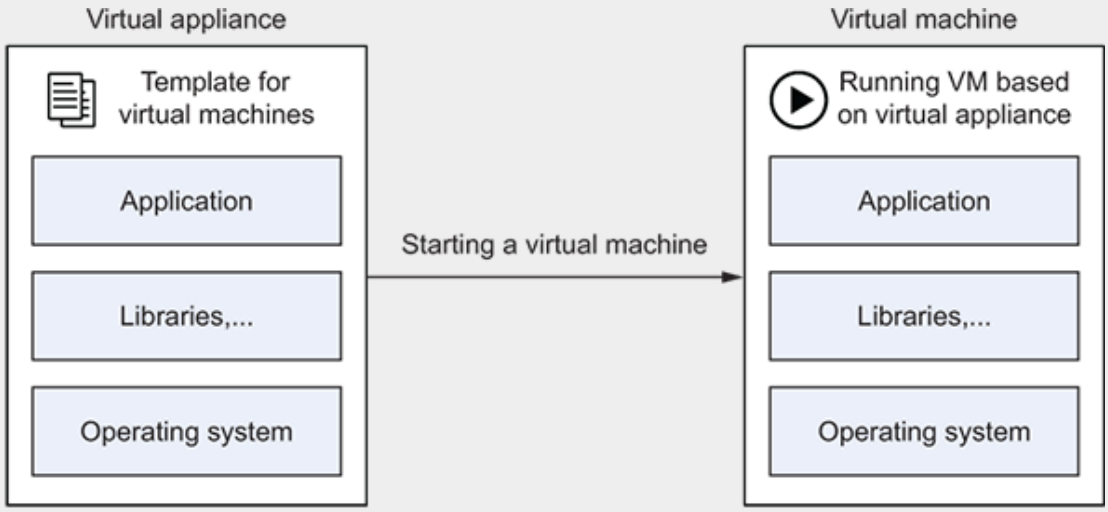
publisher. We prefer working with Amazon Linux, because it’s maintained and optimized by AWS.

## Virtual appliances on AWS

As shown in the following figure, a virtual appliance is an image of a virtual machine containing an OS and preconfigured software. Virtual appliances are used when the hypervisor starts a new VM. Because a virtual appliance contains a fixed state, every time you start a VM based on a virtual appliance, you’ll get exactly the same result. You can reproduce virtual appliances as often as needed, so you can use them to eliminate the cost of installing and configuring complex stacks of software. Virtual appliances are used by virtualization tools from VMware, Microsoft, and Oracle, and for Infrastructure as a Service (IaaS) offerings in the cloud.




**A virtual appliance contains a template for a virtual machine.**

The AMI is a special type of virtual appliance for use with the EC2 service. An AMI technically consists of a read-only filesystem including the OS, additional software, and configuration. You can also use AMIs for deploying software on AWS, but the AMI does not include the kernel of the OS. The kernel is loaded from an Amazon Kernel Image (AKI).

Nowadays, AWS uses a virtualization called Nitro. Nitro combines a KVM-based hypervisor with customized hardware (ASICs), aiming to provide a performance that is indistinguishable from bare metal machines.

The current generations of VMs on AWS use hardware-assisted virtualization. The technology is called a hardware virtual machine (HVM). A virtual machine run by an AMI